

Digital Twins for Water Supply Networks

DANSIS Seminar

2026.05.27

Klavs Høgh

M.Sc. Hydroinformatics

NIRAS A/S

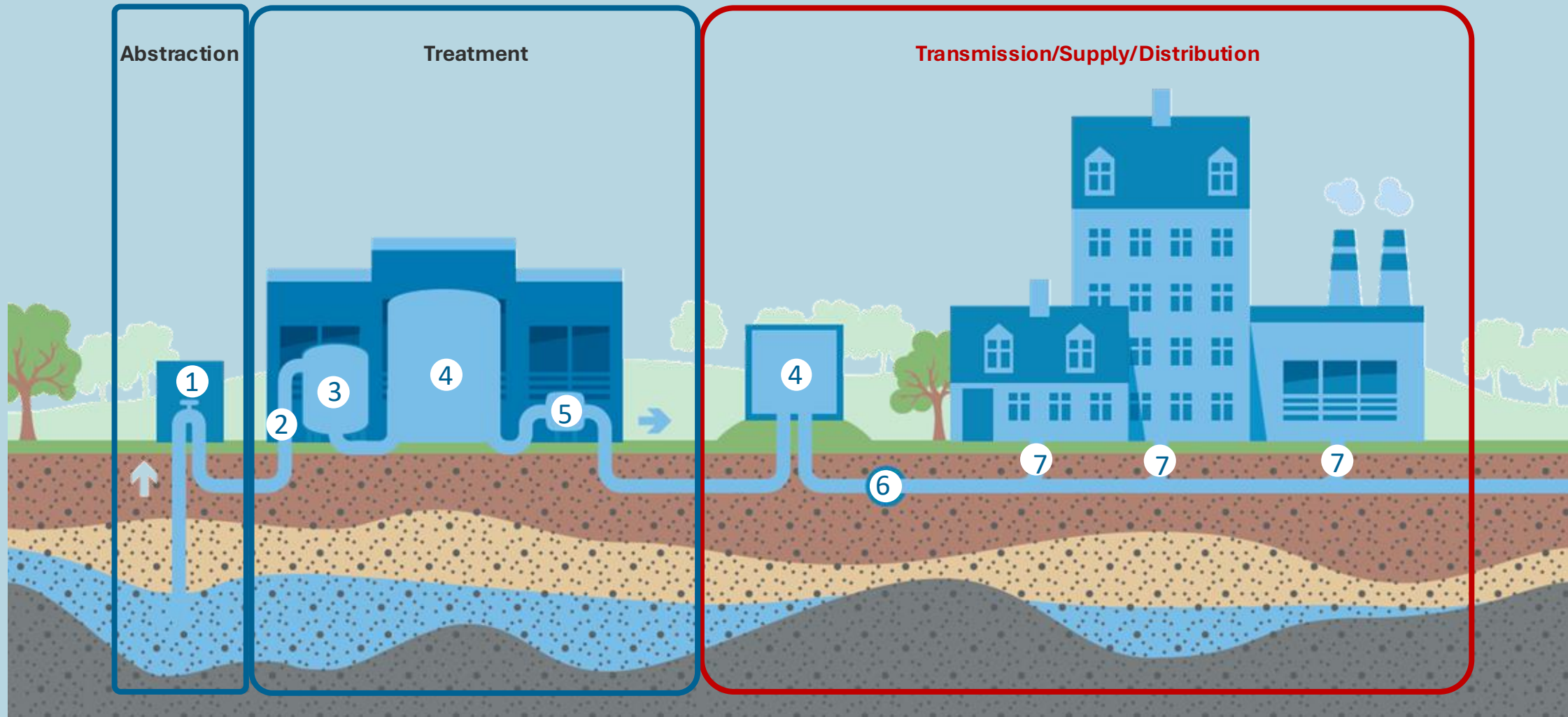
kvh@niras.dk



More than 30 years' experience with water supply systems specializing in smart water systems, digital twins, hydraulic modelling, non-revenue water (NRW) management, water loss control, energy efficiency and integration with information and communication technologies (ICT). Extensive experience with using AMR/AMI systems in drinking water systems and has been involved in several research and implementation projects developing new innovative applications that utilizes the vast amount of detailed customer information being available from the new metering systems.

The Drinking Water System

From source to tap

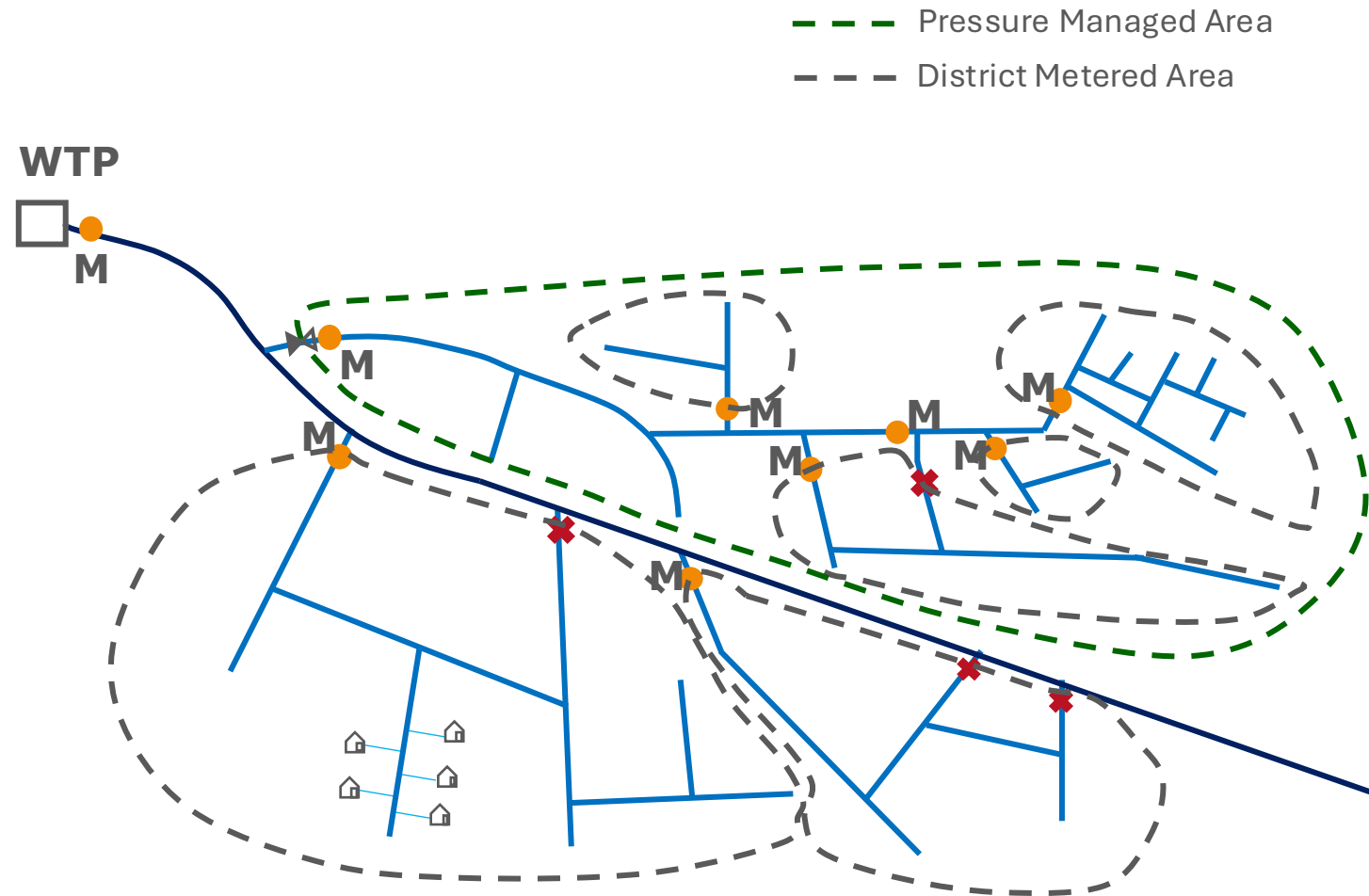


Sectioning of a Water Supply Network

Pressure Managed Areas and District Metered Areas

Dividing (sectioning) the water supply network into Pressure Managed Areas and District Metered Areas allows for efficient control, monitoring and optimization of the operation.

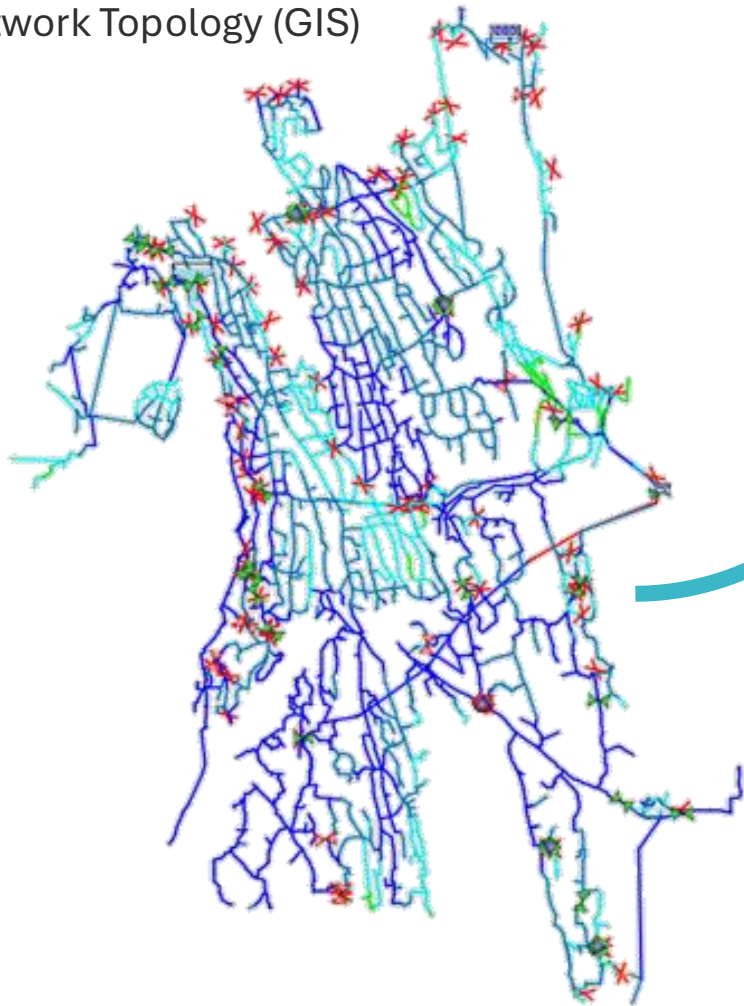
- A Pressure Management Area (PMA) is an area of the water distribution network where the pressure can be controlled independently of the remaining network.
 - PMAs are constructed by a combination of water production locations and characteristics, pressure requirements and elevation profiles.
- A District Metered Area (DMA) is a discrete area of the water distribution network where all connections to the remaining network are either closed or monitored for flow.
 - DMAs are constructed by a combination of removing pipe connections, installing boundary valves and constructing monitoring stations.



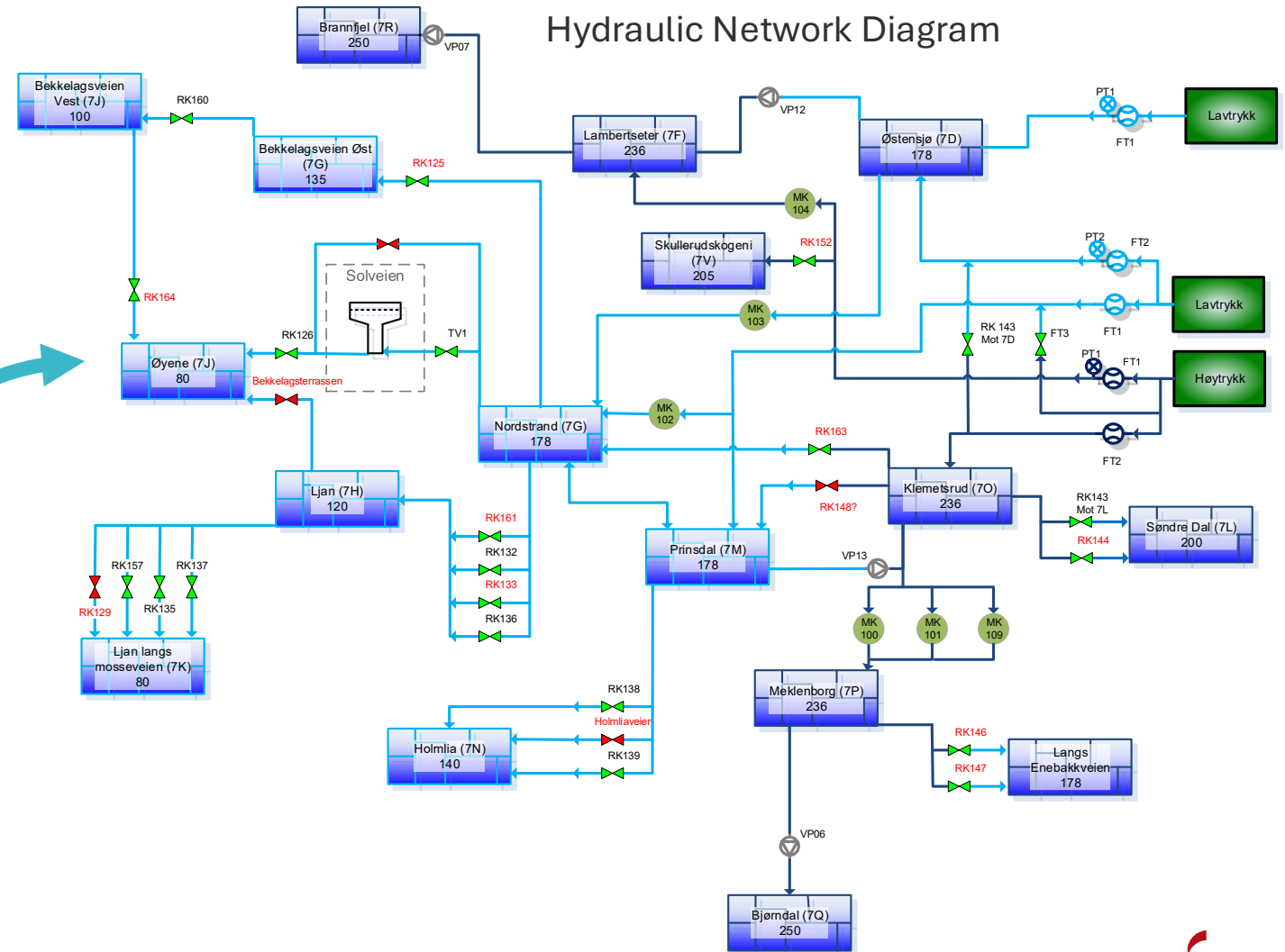
Hydraulic Network Diagrams

A PID for Water Supply Networks

Network Topology (GIS)



Hydraulic Network Diagram



The LEAKman Concept

A Danish concept for Non-Revenue Water Management & Water Loss Control



- **Key ICT systems** in a “modern” NRW Management setup.
- **The importance of open data sources** and interface options.
- **GIS, the centralized data** repository for all topographical information.
- **SCADA, the centralized data** repository for all operational measurements
- **HMS, integrate the hydraulic modelling** system with GIS, CIS and SCADA.
- **MIS, the added value** of implementing a Management Information System to monitor performance of e.g. NRW initiatives etc.

What is a Digital Twin?

In context of a water distribution system

A Digital Twin is a digital representation of the real-world water distribution system containing both static and dynamic data from all available data sources such as:

- GIS (Geographical Information System), including network topology, landbase data, building information etc.
- CIS (Customer Information System), including customer data, billing data, meter reading, meter information etc.
- AMS (Asset Management System)
- WO (Work Orders)
- WQ (Water Quality), including water quality requirements, sampling locations, sampling results etc.
- SCADA (Supervisory Control And Data Acquisition), including monitoring data, monitoring units, data quality, set-points from pumps and valves etc.
- ...and other available ICT systems containing information and data related to the operation of the distribution system.

A Digital Twin requires digitized data and digital systems and then:

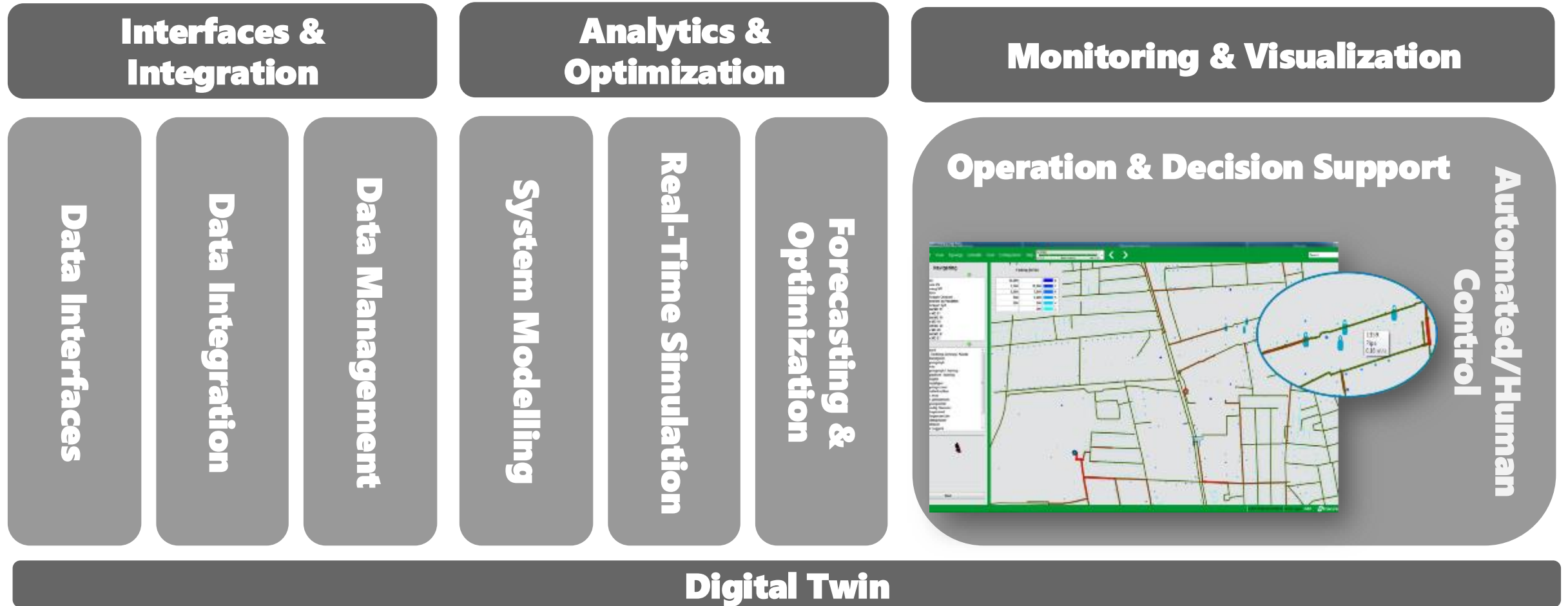
- Integrate data from the digital (ICT) systems.
- Create a common (centralized) data repository (database).
- Add data validation and data integrity to ensure high validity of the data basis.

A Digital Twin enables digital transformation to enable Shifting from Empirical Operation to Data-Driven Decision-Making through:

- Implementation of a flexible and robust analytics platform.
- Automation of management and performance monitoring and reporting, which significantly reduces manual work involved, ensures uniform reporting and reduces (eliminates) the risk of human errors.
- Online support to simulate operational changes before implementation being activate to verify effects.
- Online forecasting of operation providing early warning on future operational challenges and conflicts.

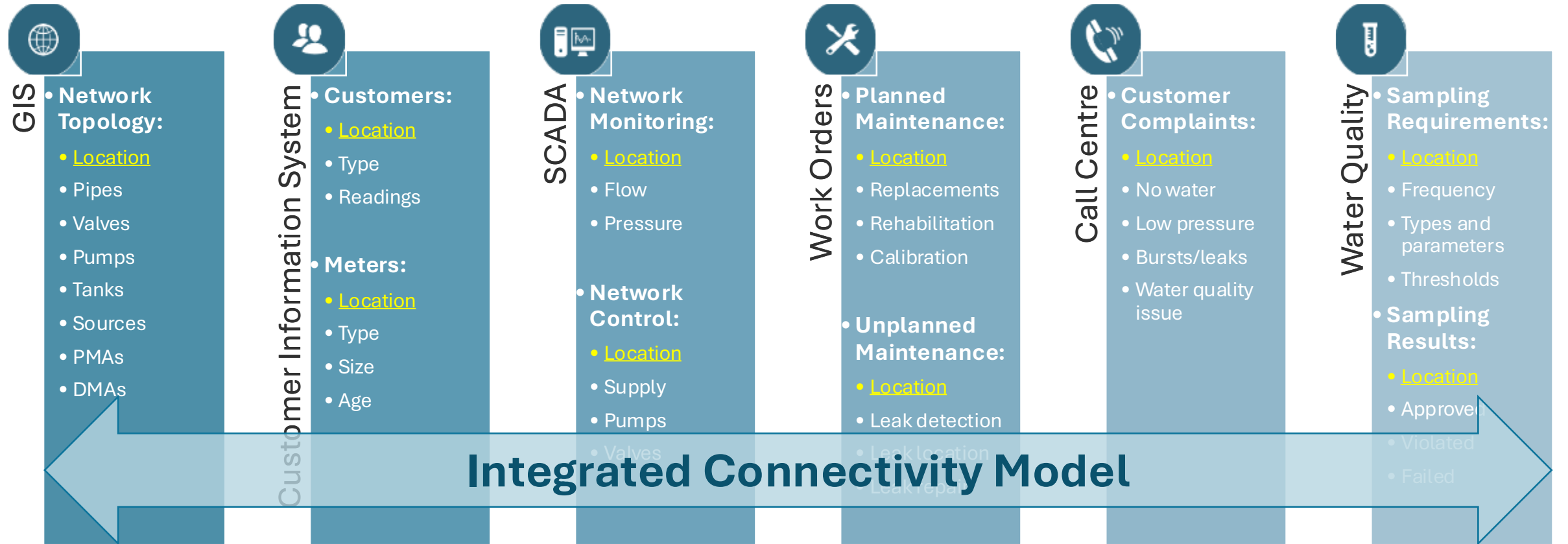
The Digital Twin

Basic architecture, layout and functionality



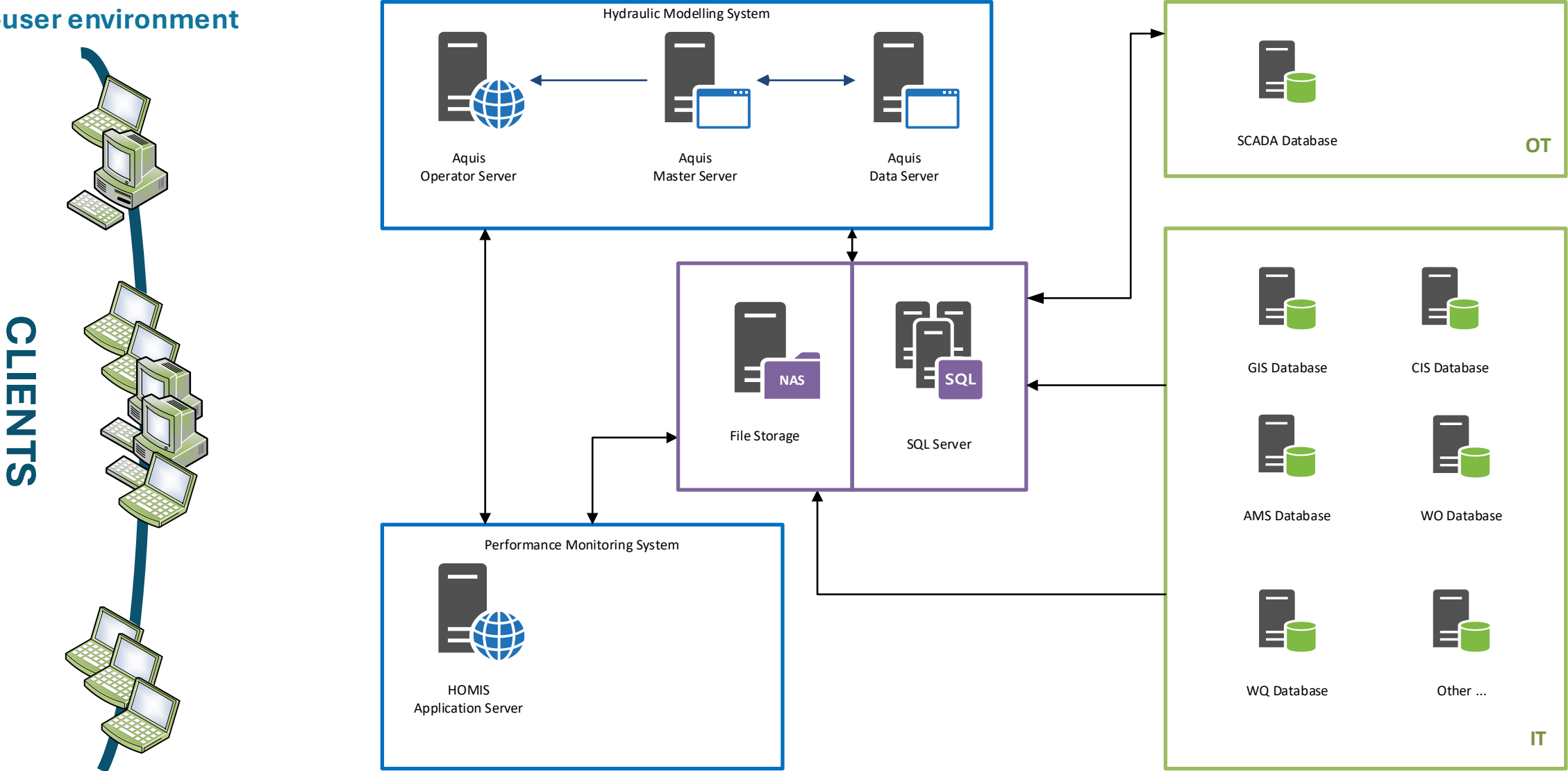
Interfaces & Integration

Individual ICT systems are often available as siloed systems



Enterprise Architecture

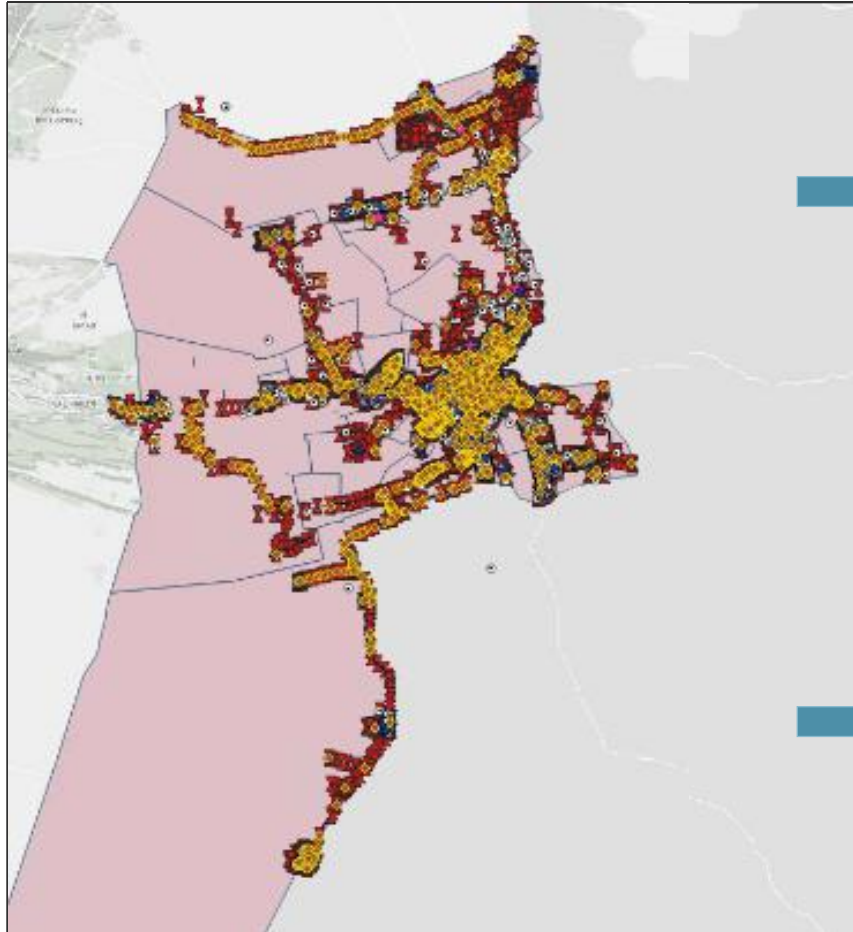
A multi-user environment



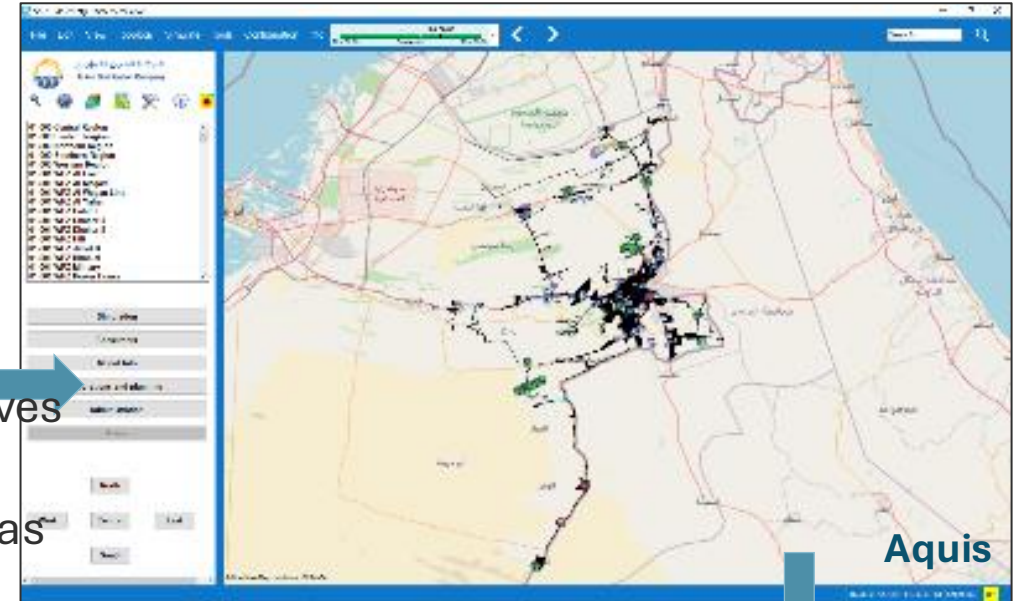
Building a Digital Twin

Real-time Hydraulic Modelling and Online Performance Monitoring

Data Sources



Main Pipes
Sector Main Pipes
Service Pipes
Isolation & Control Valves
Regions
Pressure Managed Areas
District Metered Areas
Customer Demands & Meter Information
Water Quality Sampling
Work Orders
SCADA Monitoring
Customer Complaints



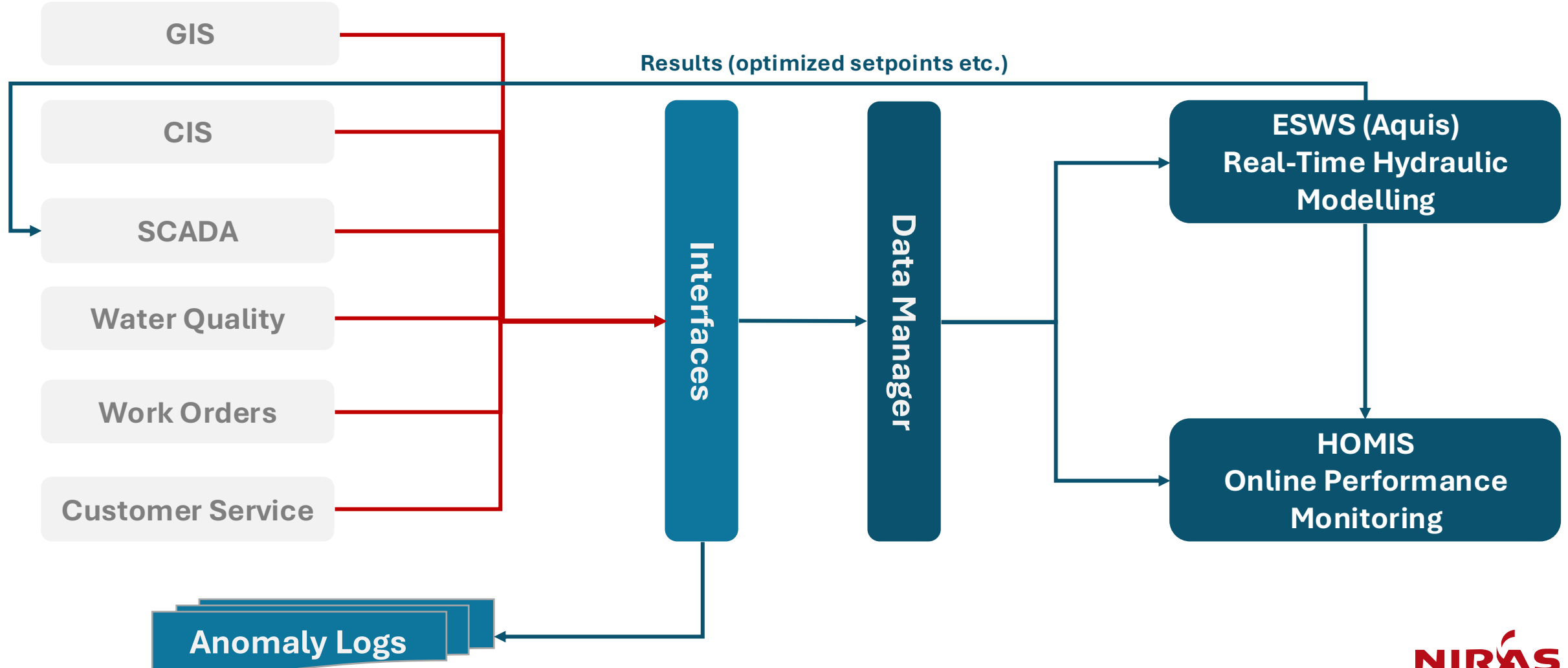
Real-Time Hydraulic Modelling



Online Performance Monitoring

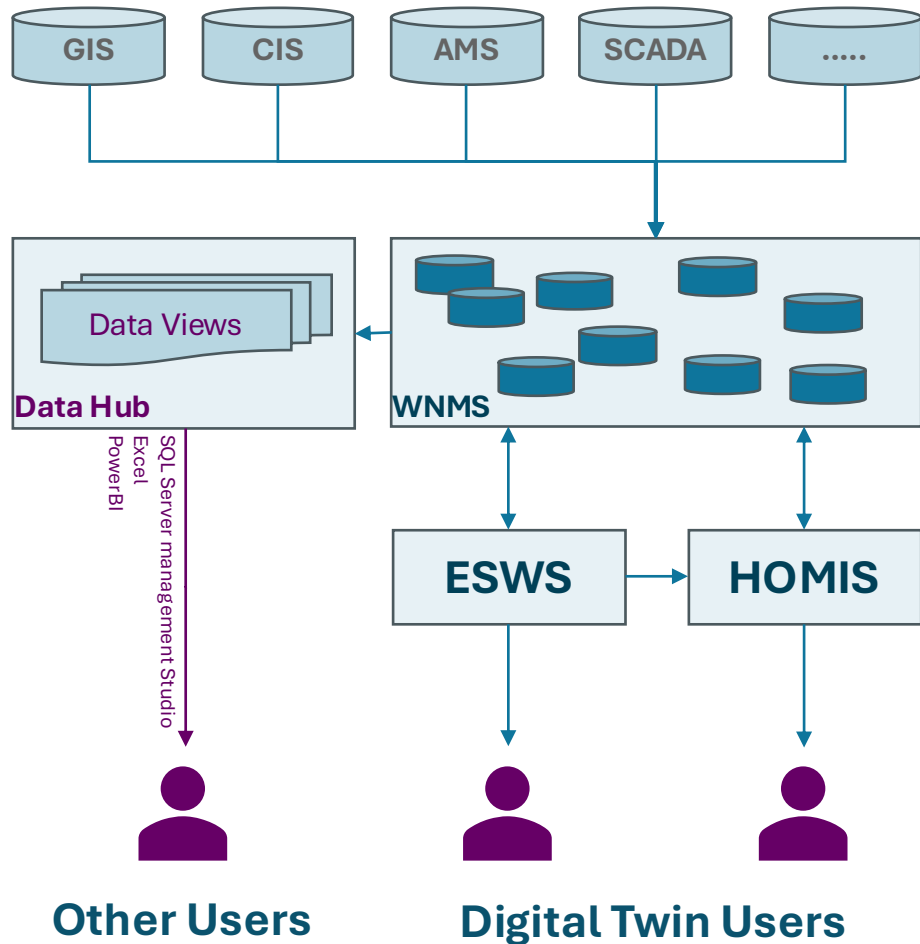
Data Flow & Interfaces

Data preprocessing, validation, integrity testing and anomaly reporting



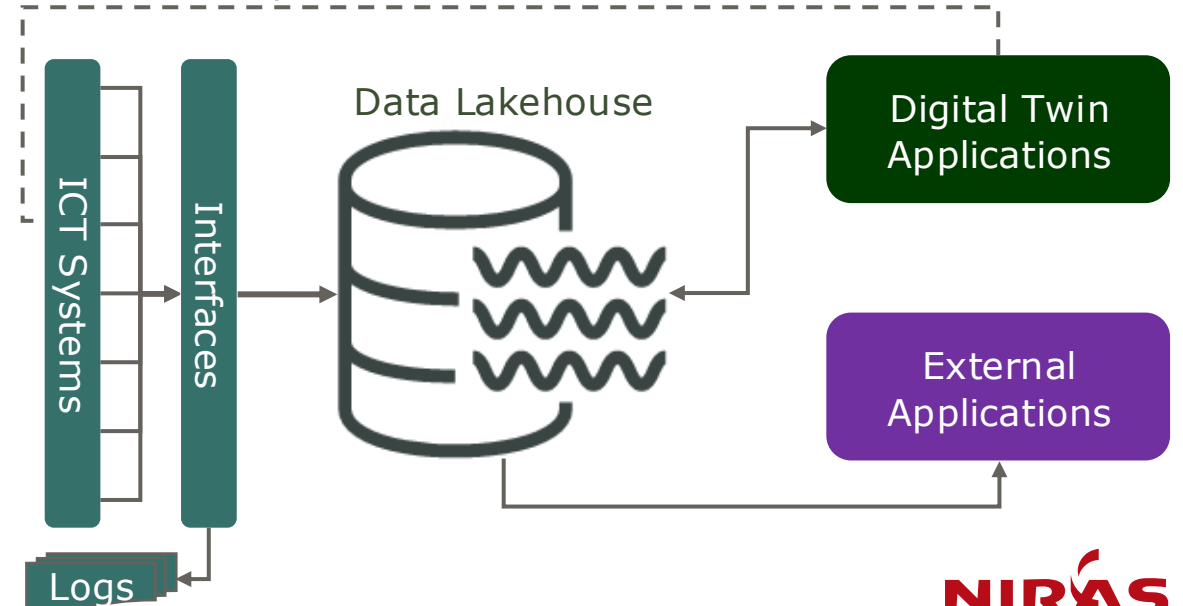
Setting Up a Data Lakehouse

Increasing the value of the Digital Twin



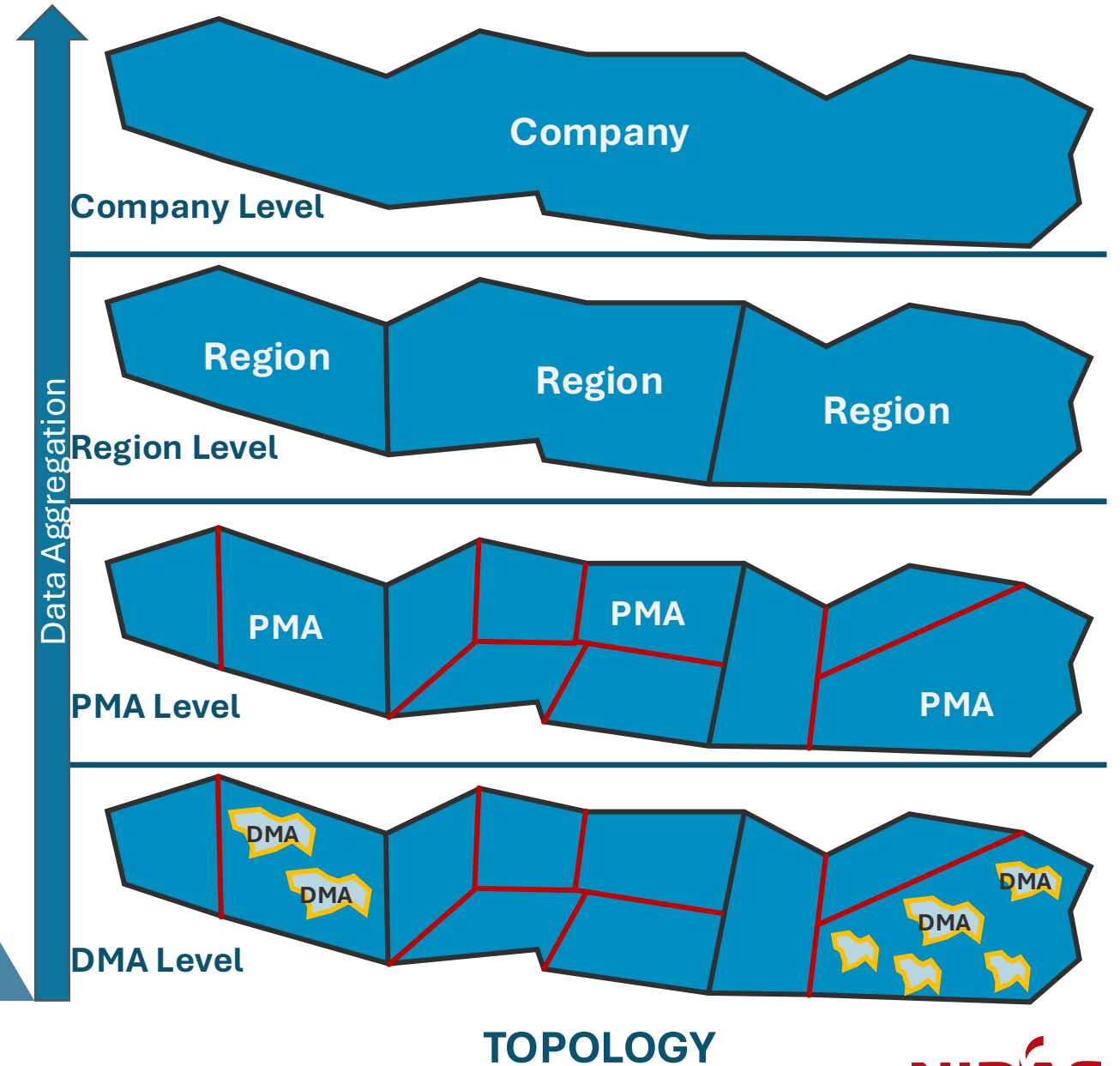
Apply a Data Lakehouse (a combination of a Data Lake and a Data Warehouse) to achieve:

- A Unified Platform containing all data received/collected, processed and an analytics results/output from the Digital Twin.
- An Open and Flexible structure publishing all quality assured data for external use “when needed”/”on request”.
- Standard and Open protocols to directly support interaction with desktop users.



User Hierarchy & Access Control

Aligned against role, responsibility and functionality

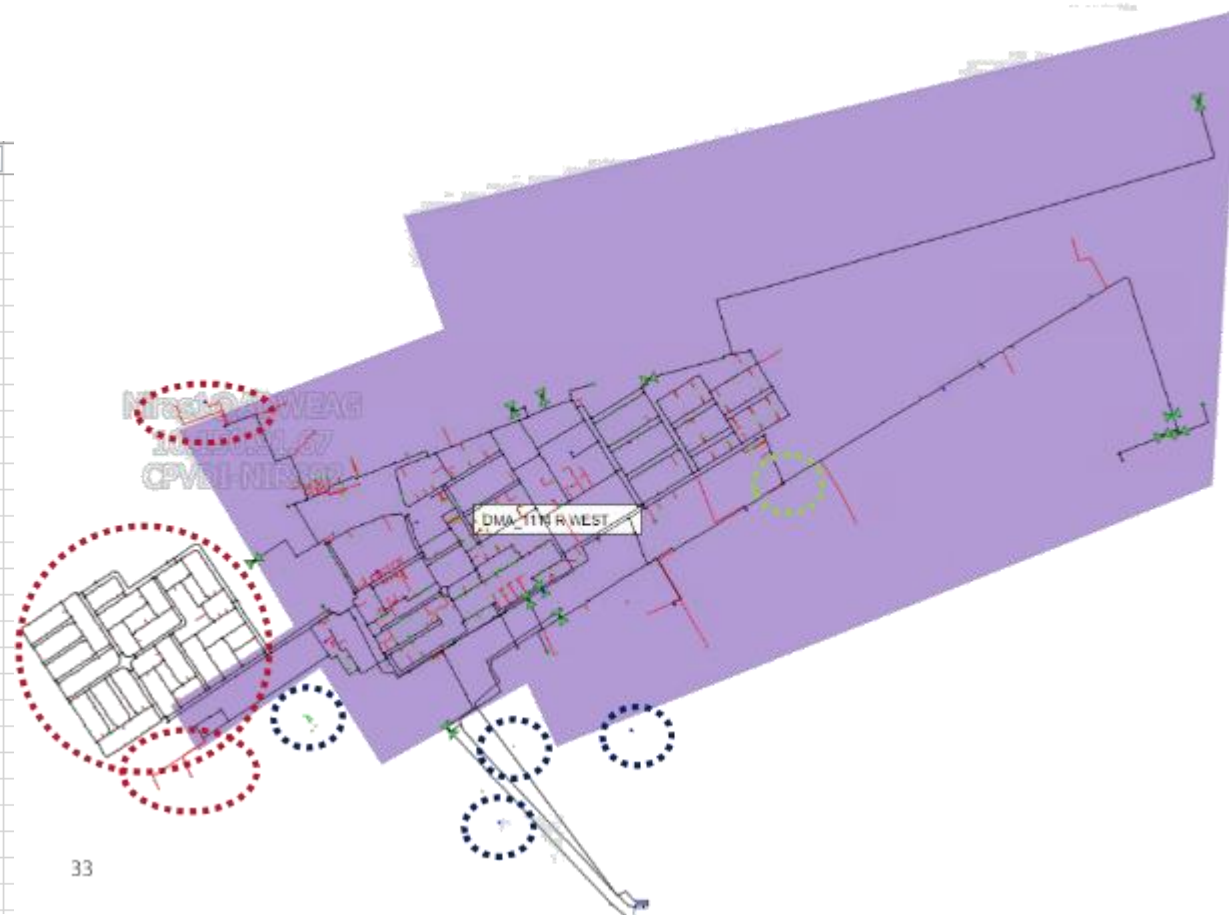


Data-Driven Improvement Initiatives – Data Consistency & Integrity

Online check across related data sources

- Check customer geocodes from CIS and verify against GIS.
- Apply connectivity model to estimate missing/faulty geocodes.
- Verify boundaries.
- Check asset names across GIS, AMS, WO etc.
- Report anomalies to data owners for implementation of improvements, corrections and rectifications.

SAID	MTRSIZE	PREMTYPE	XYINFO	RATING	DMA_PMA	YR_M3
7125999293	80	GOV	NoValidSource	8	NULL	5,677
5603627291	200	STATION	AREA	7	DMA_1202 R NORTH	57,425
6524390850	400	TFS	AREA	7	PMA Al Maqam	45,763
7186376993	100	GOV	AREA	7	PMA_NORTH_303_Al Jabib West_11/3	29,046
1373267722	200	TFS	AREA	7	DMA_1003 R SOUTHERN	27,302
9903116412	80	GOV	AREA	7	PMA Towaisa Line	22,137
0127887927	1,000	FARM	AREA	7	DMA_1205 R NORTH	21,250
0031225175		FARM	AREA	7	PMA Hili BBS	10,309
0636047367		FARM	AREA	7	PMA Hili BBS	10,309
2804713612		FARM	AREA	7	PMA Hili BBS	10,309
3868618105		FARM	AREA	7	PMA Hili BBS	10,309
5306861436		FARM	AREA	7	PMA Hili BBS	10,309
7942741295		FARM	AREA	7	PMA Hili BBS	10,309
4402918852	25	PALACE	AREA	7	DMA_204 R DAHMA	10,084
1320396749	100	TFS	AREA	7	DMA_1302 R UM GHAFFA	9,507
3383010817	40	FARM	AREA	7	DMA_1205 R NORTH	6,924
3834810615	50	FARM	AREA	7	PMA Hili BBS	6,759
1231310879	50	FARM	AREA	7	DMA_1205 R NORTH	6,175
1173400900	50	GOV	AREA	7	PMA Um Ghafa	5,736
1519210686	40	VILLA	AREA	7	DMA_807 R ZAKHER	5,199
0304582829		FARM	AREA	7	PMA Hili	4,337
6223800461	40	FARM	AREA	7	DMA_1205 R NORTH	3,088
3507296210	200	STATION	AREA	7	DMA_701 R POWERHOUSE	2,648
4676542278	150	STATION	AREA	7	DMA_701 R POWERHOUSE	2,648
2181531990	40	GCUN	AREA	7	PMA Al Maqam	2,331
0776737837	25	PARK	AREA	7	DMA_201 R DAHMA	2,303
3176020018	50	CAMP	AREA	7	DMA_402 R KHABISI	2,262
8806470744	50	VILLA	AREA	7	DMA_808 R ZAKHER	2,045

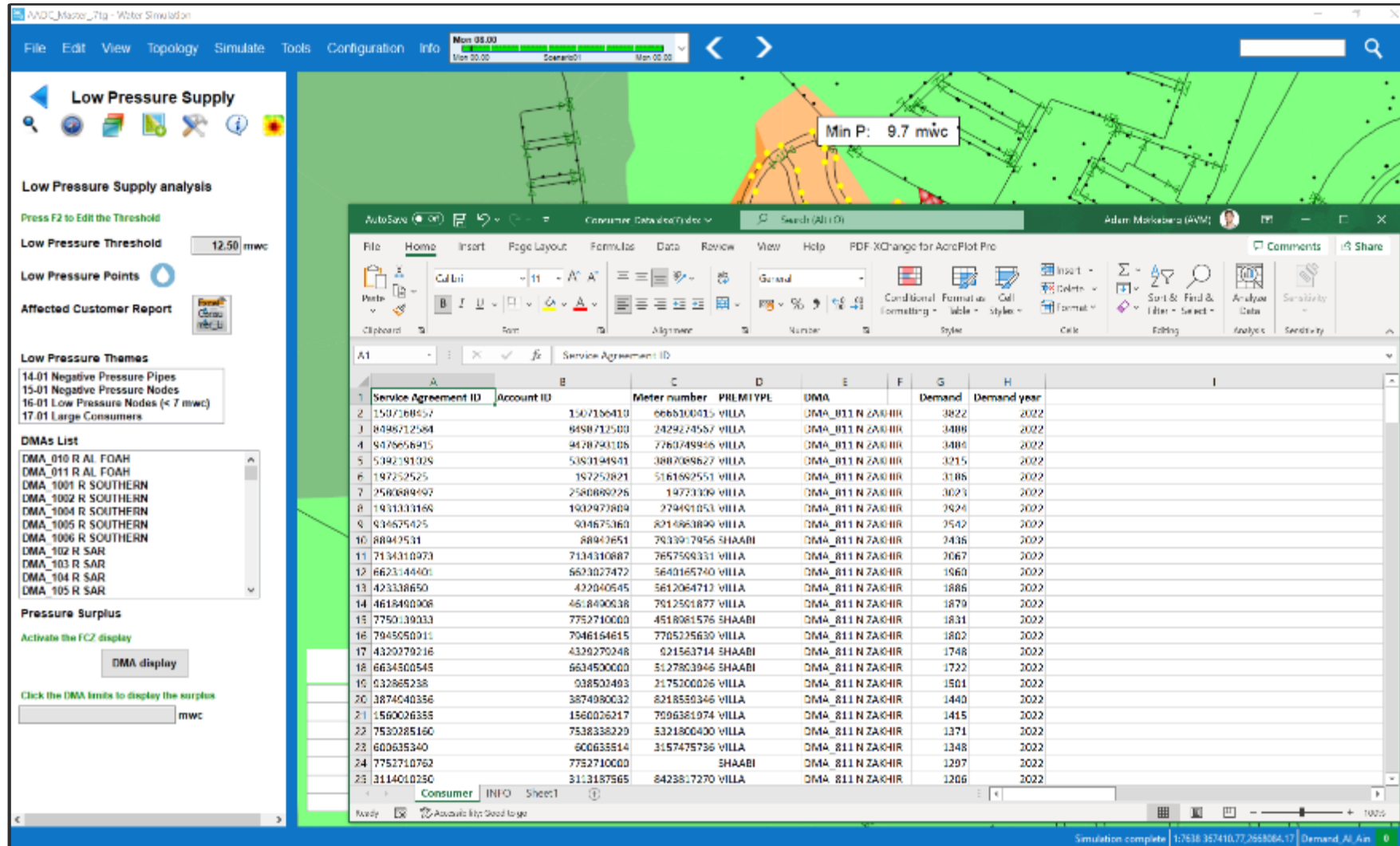




Pressure Management & Energy Optimization

Identification of Low-Pressure Supply Areas

Real-Time Hydraulic Modelling

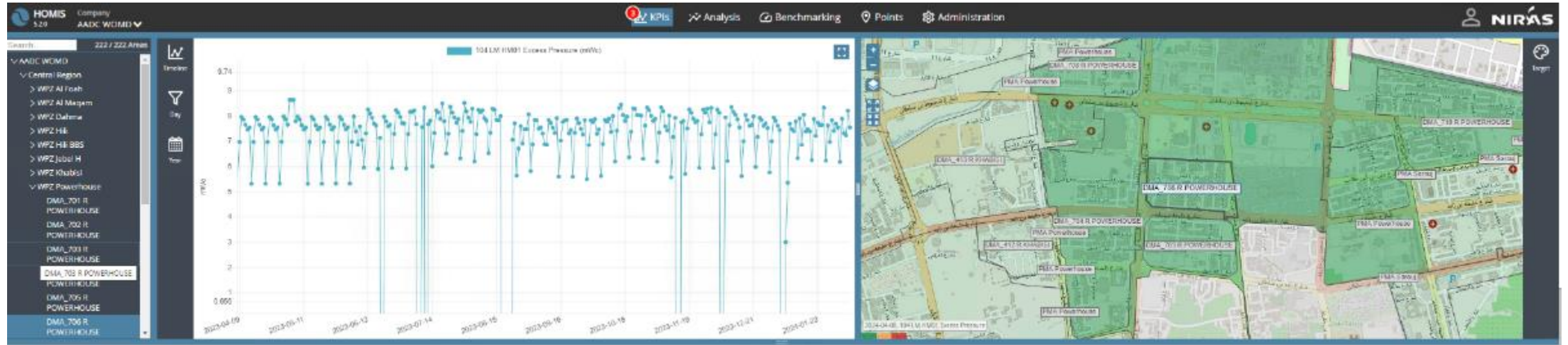


Real-Time Hydraulic Modelling:

- Using ESWS (Aquis) by Schneider Electric.
- Online connection to SCADA and CIS.
- Simulation cycle update every 30 minutes.
- Service level at 1.25 bar

Identification of Low-Pressure Supply Areas

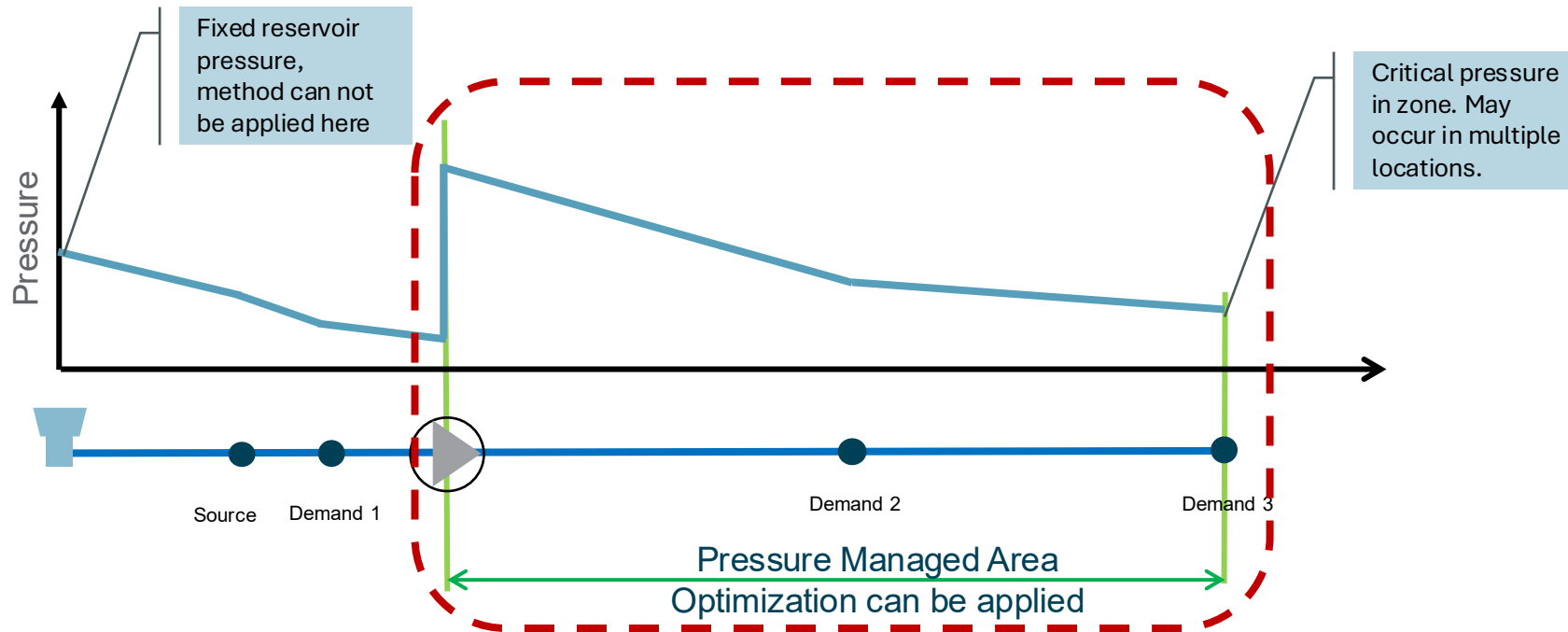
Real-Time Hydraulic Modelling



- Bi-directional interface with SCADA!
- The “Required Supply Pressure” is sent back to SCADA as a reference value, which the Operators can use to adjust the Set-Point of the DMA inlet valve control.

Pressure Optimization

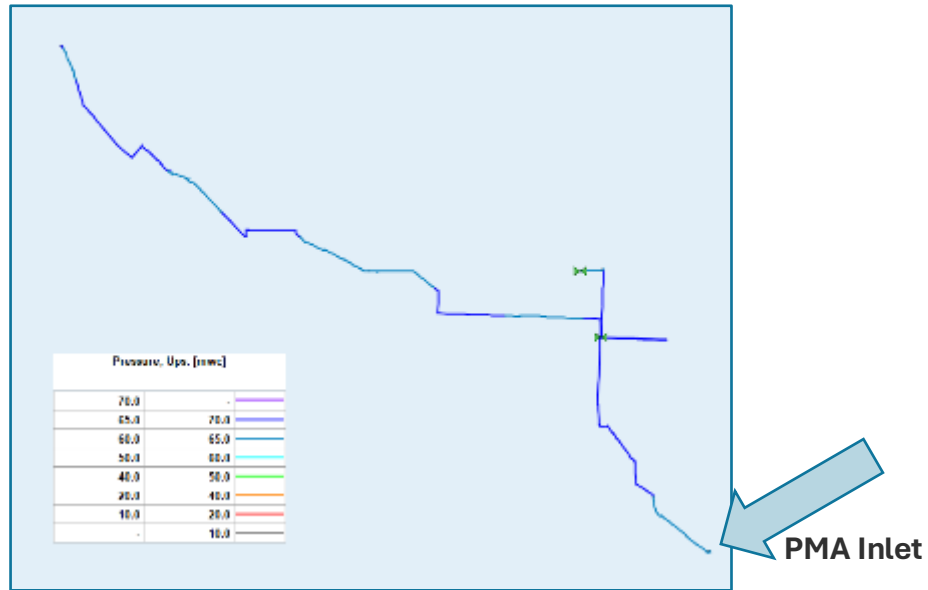
Working on zones (PMAs or DMAs) in a distribution system



- Secures that the pressure at the critical point is always maintained at (or above) the pressure service level.
- The (online) hydraulic model calculates the required pressure at the zone inlet and returns the optimized pressure setpoint to SCADA.
- Optimization will consider any maximum pressure constraints.
- Optimization will run e.g. every 5-10 minutes and calculate/forecast optimized setpoints for the next 24 hours.
- Build up Q-P reference control relation to be used for fallback operation.

Pressure Optimization – Feasibility Study using a Digital Twin

For a Pressure Managed Area

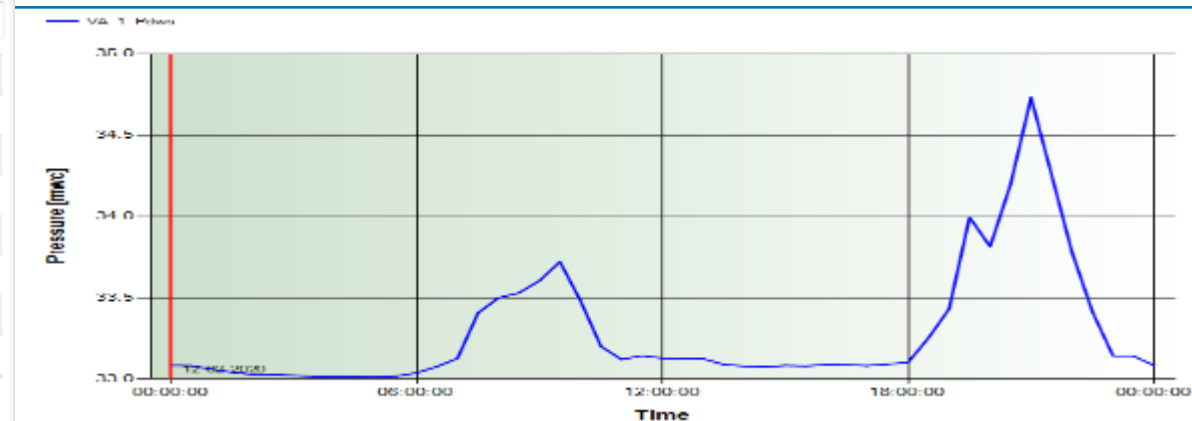
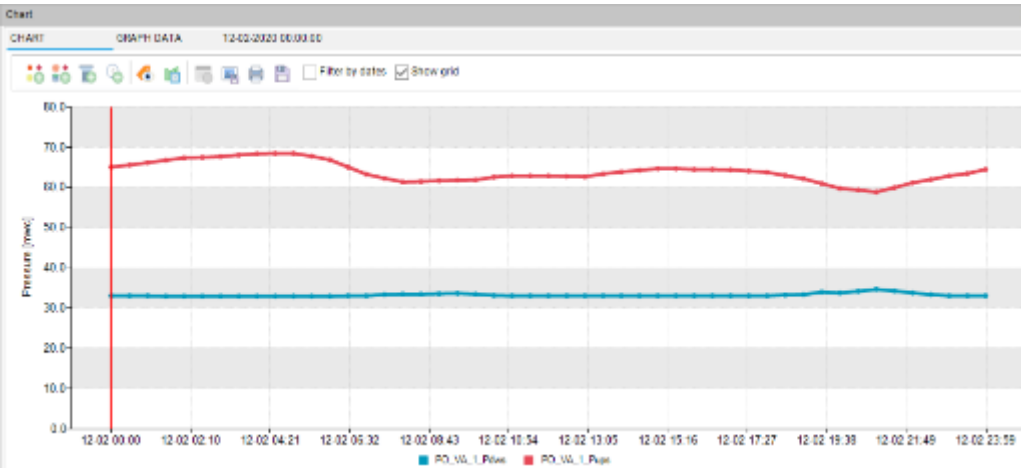


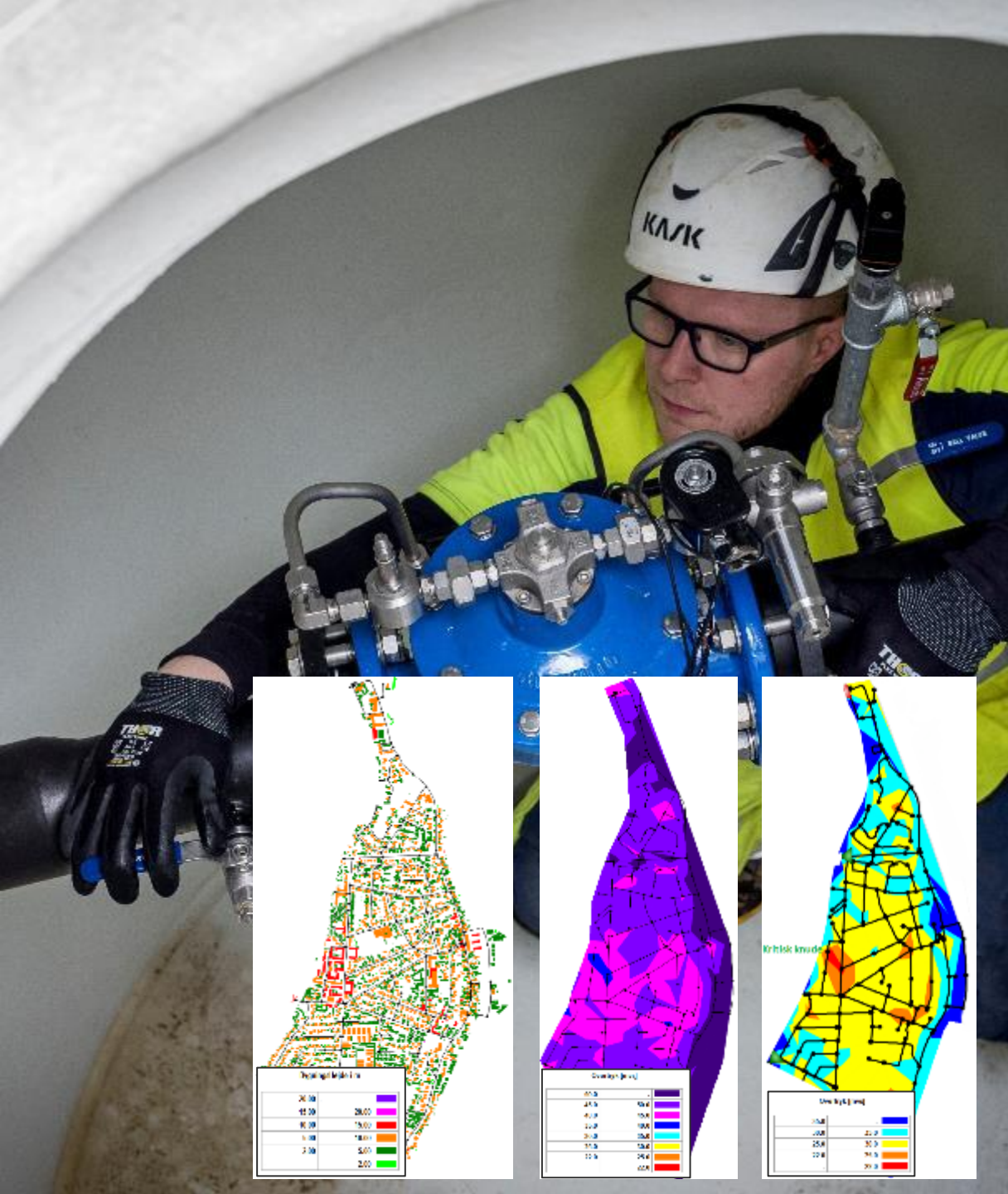
Baseline:

- The PMA is supplied from the Transmission system, with a supply pressure varying between 61-69 mwc.
- The average pressure inside the PMA varies between 59-70 mwc, with maximum peak at 75 mwc.
- Flow average flow supplied to the PMA is approximately 1,400 m³/h, and Minimum Night Flow at 330 m³/h.

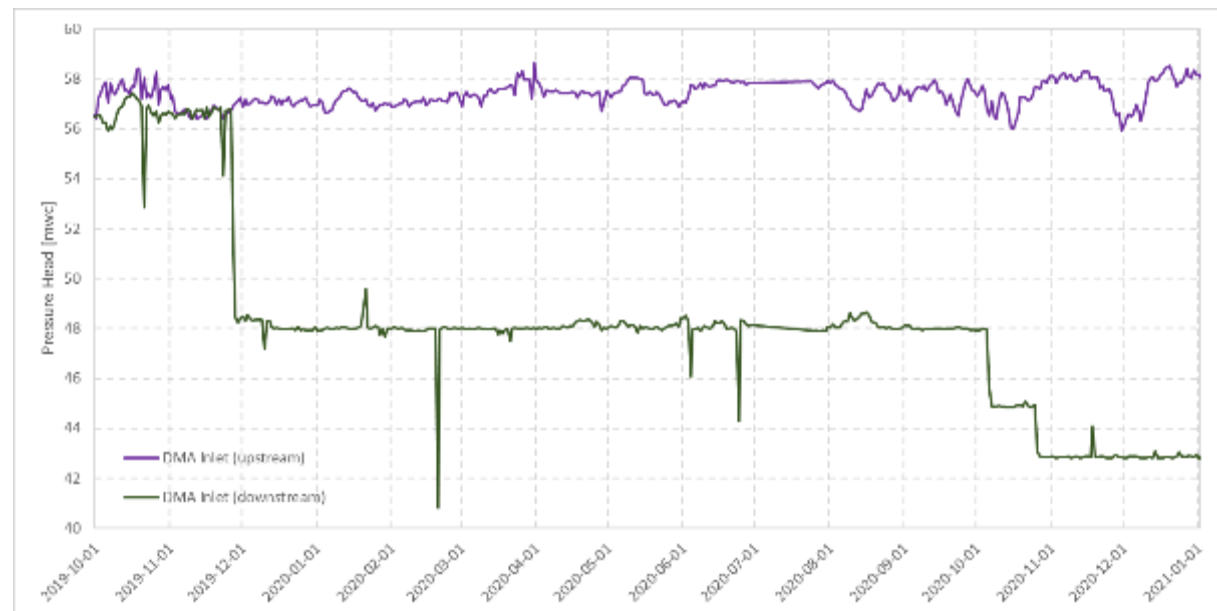
Potential of Pressure Optimization:

- The inlet pressure will vary between 37-41 mwc.
- The average pressure in the PMA will vary between 33-35 mwc with maximum at 41 mwc.
- The reduction in average pressure of 48%, will lead to direct savings in background leakage and burst rates and indirect savings in energy usage.





Pressure Optimization Novafos Denmark

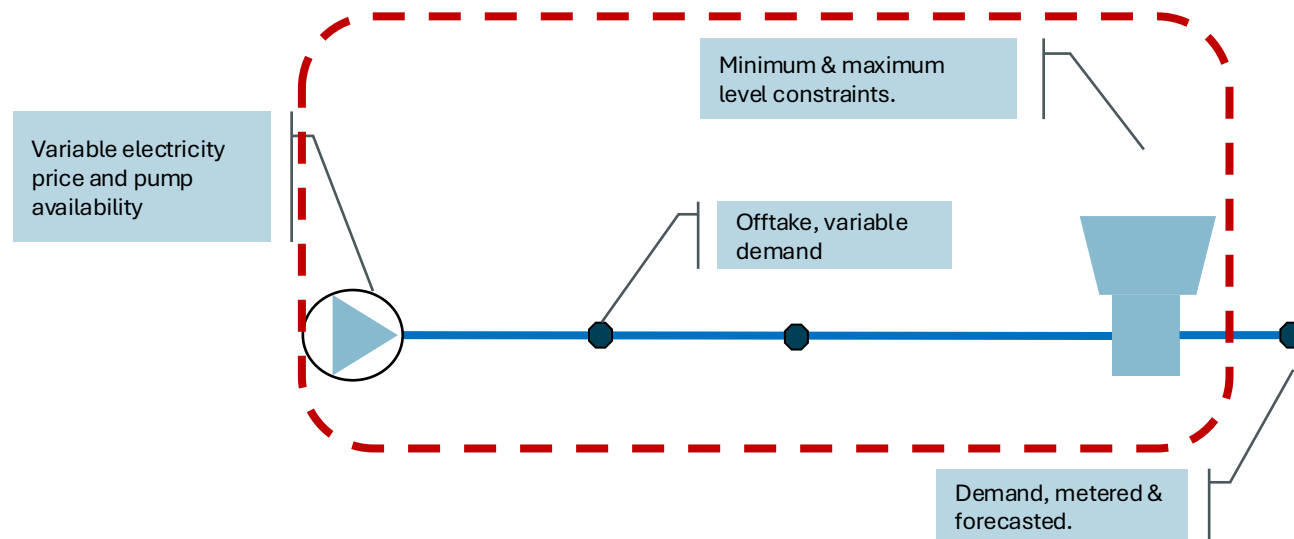


Implementation of intelligent and dynamic pressure management in a DMA:

- Designed used a Digital Twin.
- Average pressure reduced by 29%.
- Annual savings of EUR 30,000-40,000 due to reduced background leakage and reduced burst rate.

Pump-Reservoir Optimization

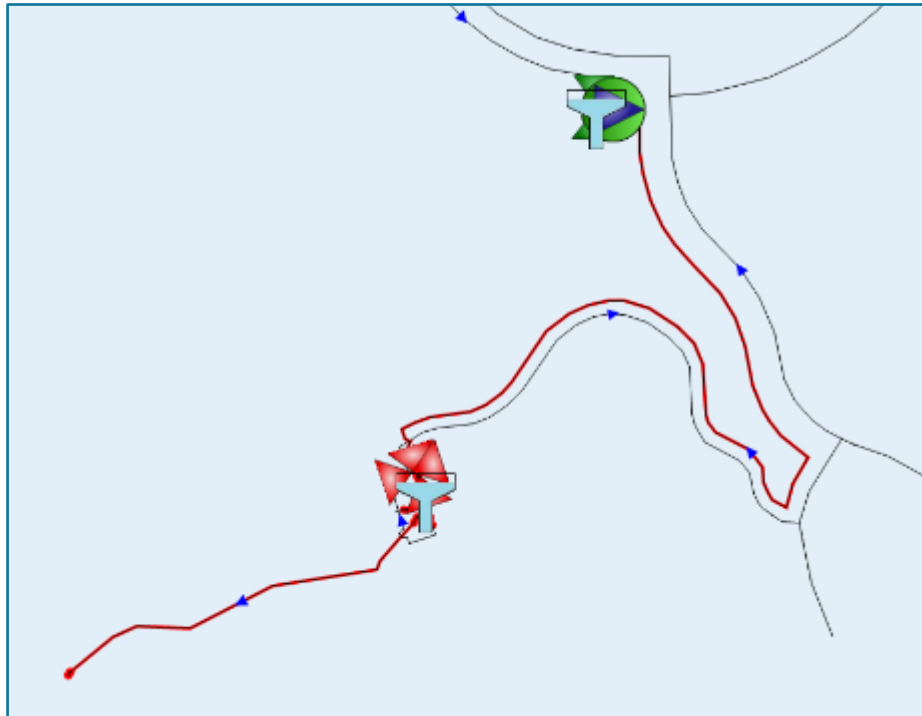
Working on pump-reservoir segments of a transmission system



- Secures that the reservoir level is maintained at sufficient level to supply the downstream demand.
- The online hydraulic model calculates the optimized pumping schedule depending on varying electricity cost, pump efficiency and frictional losses (a Non-linear optimization problem).
- Optimization will consider any maximum pressure constraints in the transmission pipeline.
- The online hydraulic model calculates the required flow at the pumping station and returns the optimized flow setpoint to SCADA.
- Optimization will run e.g. every 5-10 minutes and calculate/forecast optimized setpoints for the next 24 hours.

Pump-Reservoir – Feasibility Study using a Digital Twin

For a transmission pipeline section

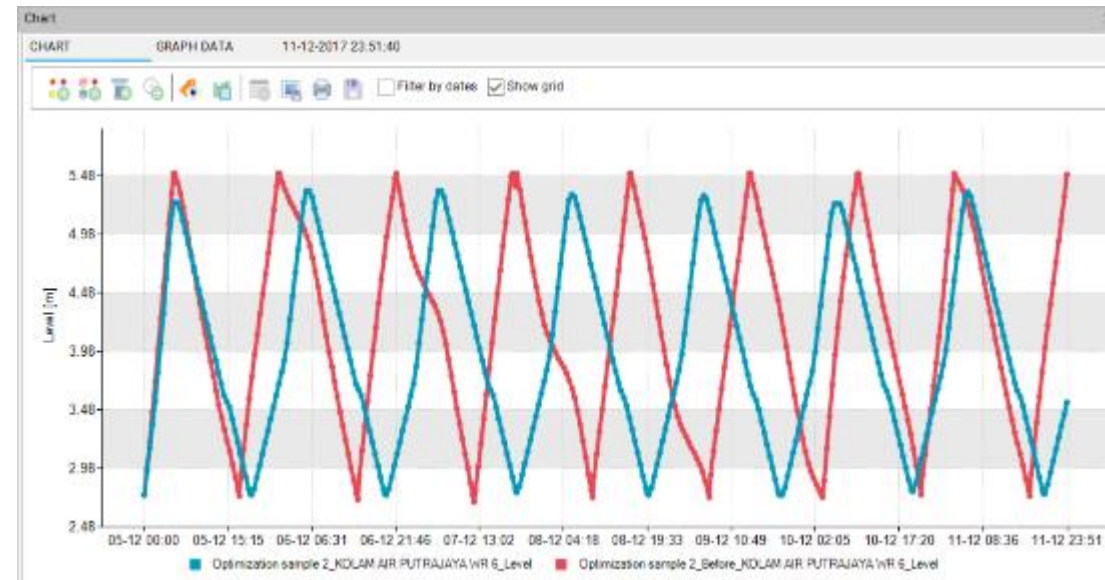


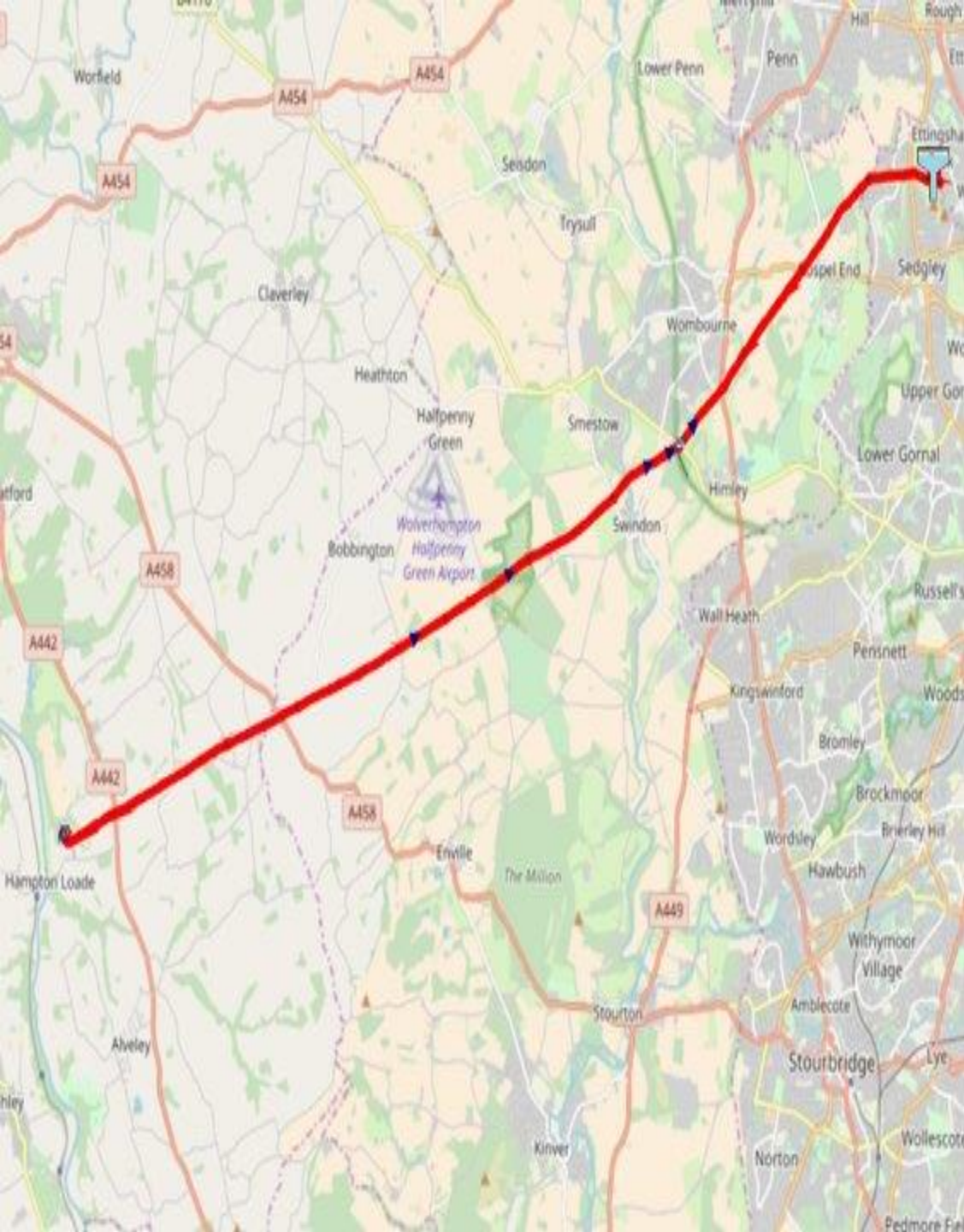
Baseline:

- Demand downstream the reservoir varies between 50-250 m³/h.
- The distance from the pumping station to the reservoir is 1.1 km (short).
- Total energy consumption is 6,592 kWh/week.

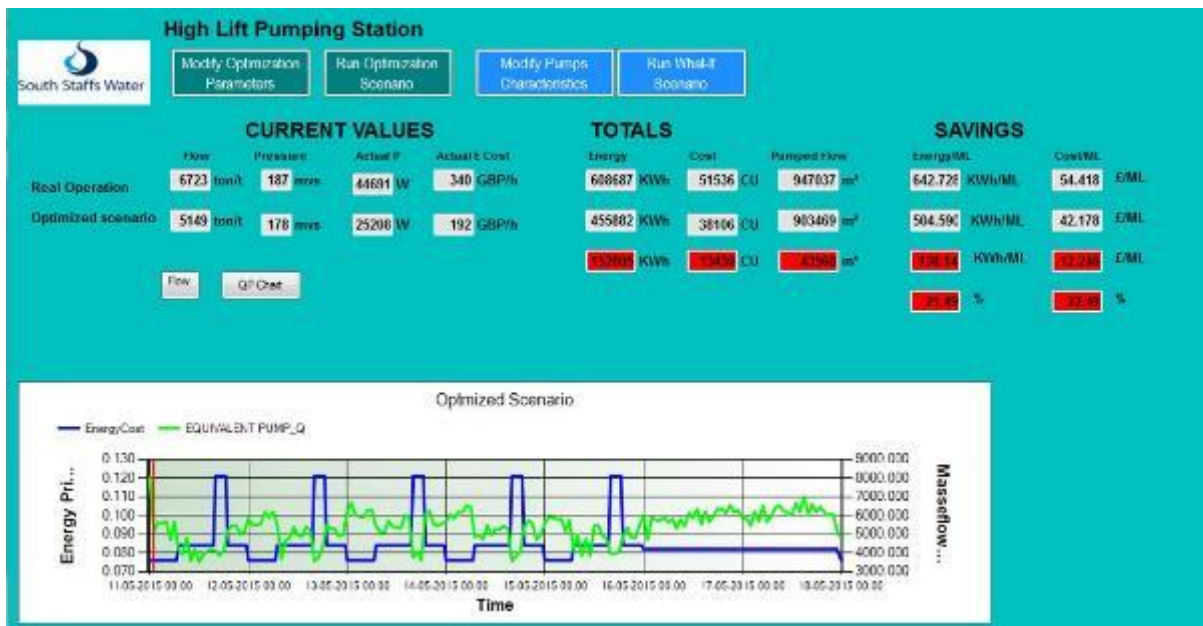
Potential of Pump-Reservoir Optimization:

- The pumps will run at lower speed for longer periods.
- Total energy consumption will be 5,240 kWh/week.
- The reduction in energy consumption will amount to 70,300 kWh/year.





Pump-Reservoir Optimization Hampton Loade U.K.



Implementation of optimized pumping operation on the transmission system:

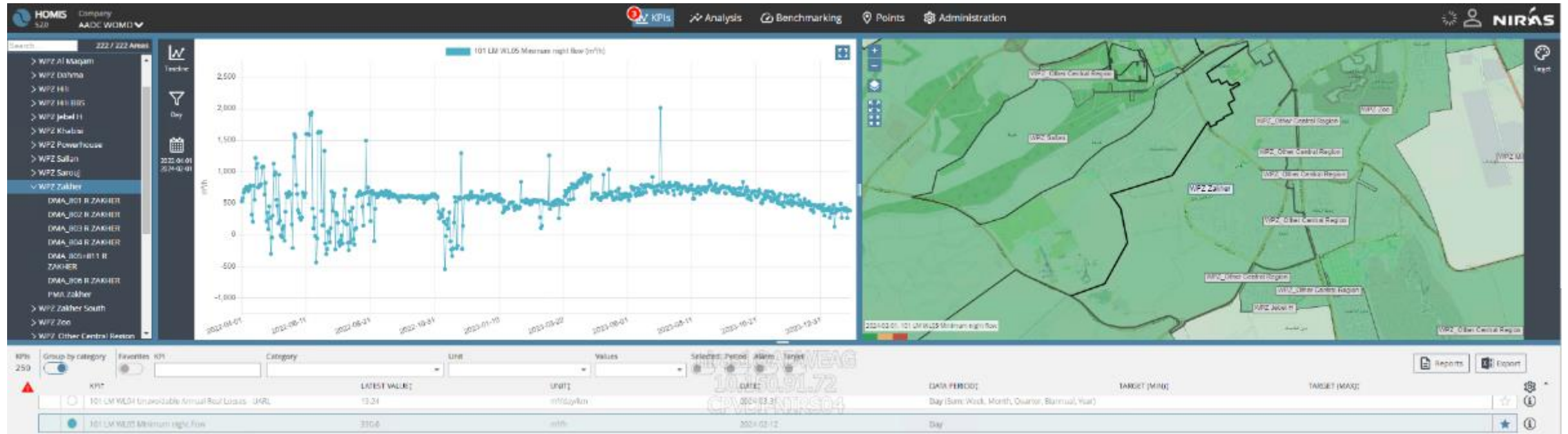
- Designed used a Digital Twin.
- Average energy consumption reduced by 21.5% for the 20 km pump-reservoir dispatch pipeline segment.



Non-Revenue Water Management & Water Loss Control

Data-Driven Improvement Initiatives – Minimum Night Flow

Online monitoring of MNF – Creating a stable and robust data flow to support Active Leakage Control



Large variation/oscillation in daily values of Minimum Night Flow have been used to first:

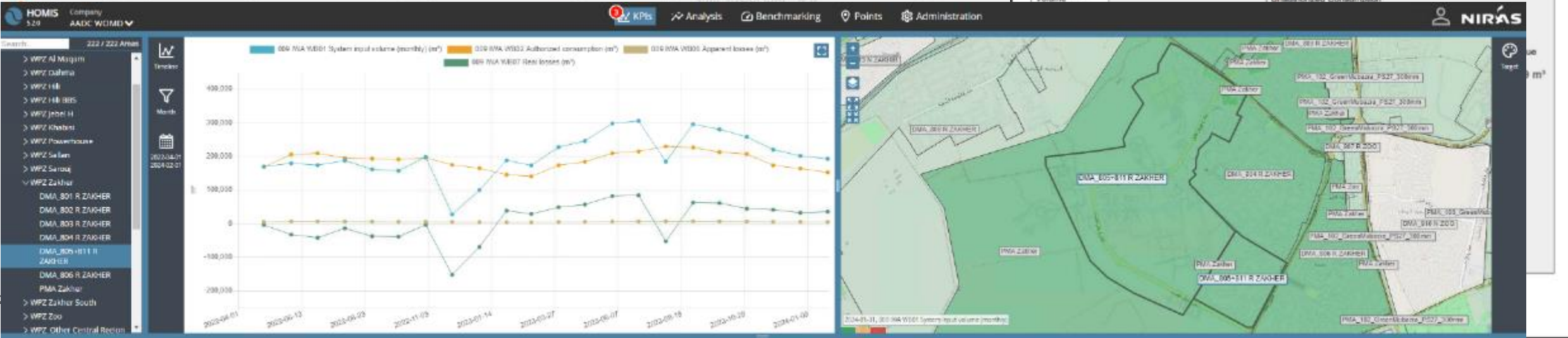
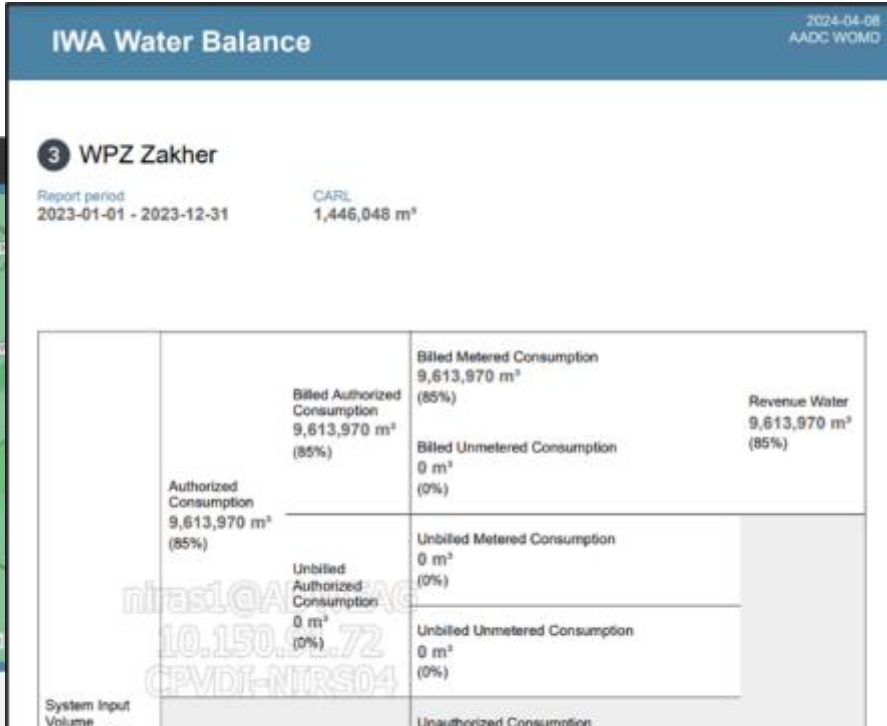
- Find errors and anomalies in SCADA configuration.
- Improve data communication with monitoring locations.
- Verify PMA and DMA boundary (emergency) valves and correct valve status and/or repair/replace defect valves.

Then:

- Used for Active Leakage Control by monitoring changes and trends in MNF.

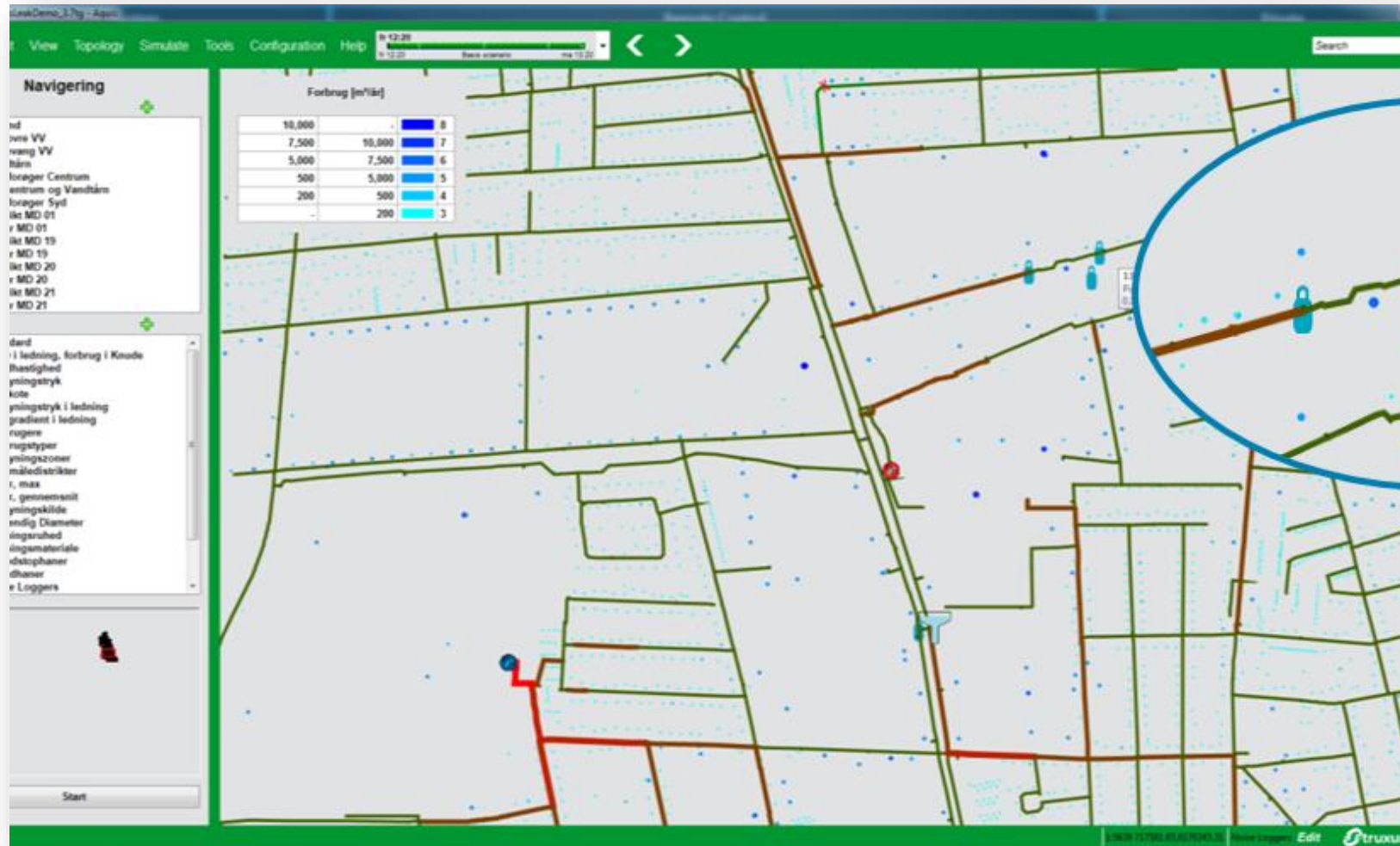
Automatic Water Balance Calculation

Monthly reporting for each PMA and DMA



Integrating Noise Loggers for Online Leak Detection

Reducing Awareness Time and prioritization of alarms



Prioritize leak alarms:

- On critical pipes.
- Near critical customers.
- At locations with critical consequences of pipe bursts.

Adding IoT Sensors to for Additional Network Monitoring

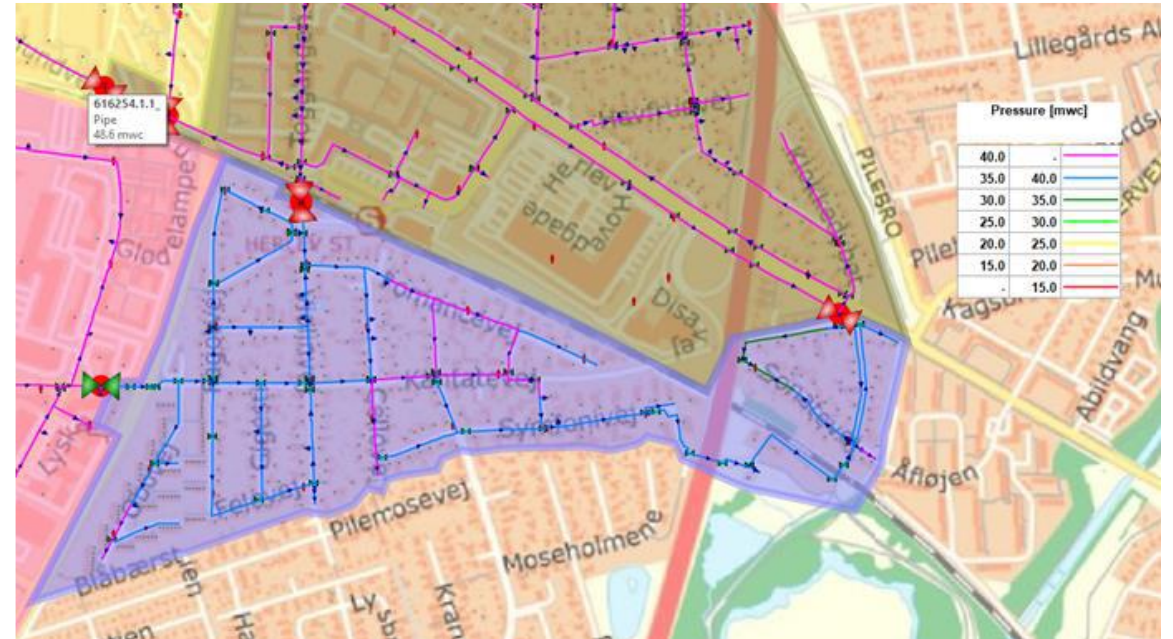
“Smart Devices” - VIDI Sensors by AVK

The **VIDI Positioner** Monitors operation of DMA boundary valves:

- Transmits data when the positioner is rotated, and a 24-hourly alive-signal.
- Monitoring operation/movement of the valve.
- Provide alarms in case of illegal use.
- **Valuable input for NRW Management!**



In use at HOFOR (Greater Copenhagen)



IoT data integrated with the Digital Twin to:

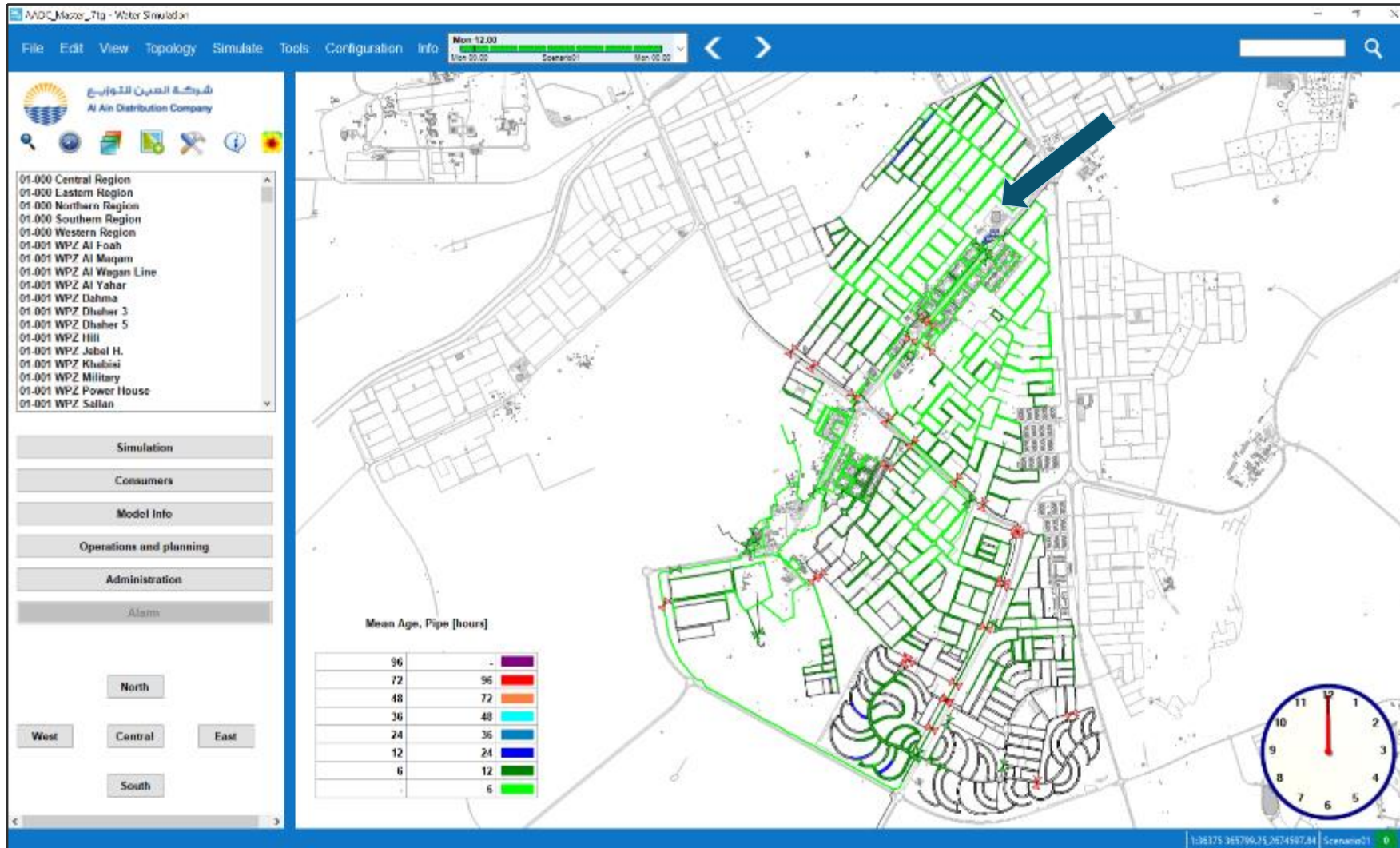
- Automatically update the valve status in the hydraulic model.
- Automatically suppress water balance calculation for the DMA when the boundary is open.



Online/Real-Time Operational Support

Real-Time Water Quality Monitoring

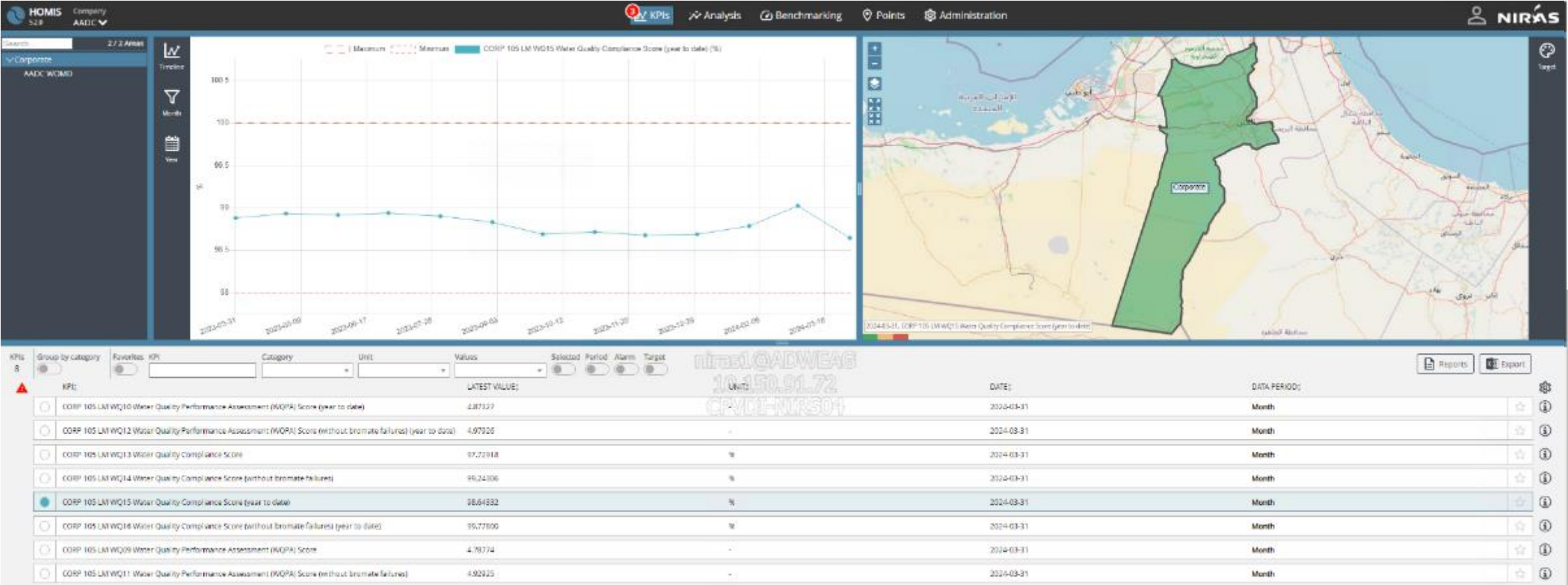
Water propagation and water age



- Water Quality simulation is automatically conducted at the end of every simulation cycle.
- Source Propagation and Water Age are always simulated by default.
- Average Water Age and Maximum Water Age sent to SCADA as supporting information for monitoring of disinfection level.

Corporate Performance Indicators

On Company Level – for high level management



Monthly Water Quality Compliance Score

(Monthly) Year-to-Date Water Quality Compliance Score



Q



A

Questions?